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ECO 442

Problem Set 5 – Answer Key

1) Congratulations! You have just been appointed the governor of the Banco Central de Costa Rica. For all of the following questions, assume the long-run (i.e., prices are fully flexible and PPP holds).

a) You have decided that you want to **decrease prices** in Costa Rica in the **long run** Would you increase or decrease the supply of Costa Rican colóns? To explain, write down the equation for the Costa Rican money market and demonstrate how the change in money supply affects the price level in Costa Rica.

b) Write down PPP between the Colón/USD exchange rate and price levels in both Costa Rica and the U.S. Illustrate the effect of a fall in the Costa Rican price level and explain what will happen to the exchange rate of the colón in the **long-run** (i.e., will it appreciate or depreciate versus the dollar?).

c) Explain the impact of this money supply change on the affordability to a Costa Rican consumer of a car produced in Costa Rica in the **long-run**. Discuss in terms of (1) the price of the car in Costa Rica (in colons), (2) the financing the Costa Rican consumer can receive, (3) the *nominal* income of the Costa Rican consumer and (4) the *real* income of the Costa Rican consumer.

d) Explain the impact of this money supply change on the affordability to a U.S. consumer of a car produced in Costa Rica in the **long-run**. Discuss in terms of (1) the price of the car in the U.S. (in dollars), (2) the financing the United States consumer can receive, (3) the *nominal* income of the U.S. consumer and (4) the *real* income of the U.S. consumer.

2) Consider two countries: Japan and South Korea. Currently, Japan has slow output growth of 1% while South Korea has robust output growth of 6%. Meanwhile, the Bank of Japan is allowing the money supply to grow by 4% a year while the Bank of Korea is choosing a money supply growth rate of 7% a year. Assume PPP holds.

a) What is the inflation rate in each country?

b) Which currency is depreciating and at what annual rate?

c) Suppose the Bank of Korea wanted to fix inflation at 2% a year. What rate of growth of the money supply would you suggest they target? Assuming nothing changes in Japan – explain how this change in the growth rate of the money supply, from the previous 7%, would affect (i) real interest rates in Korea, (ii) nominal interest rates in Korea and (iii) the exchange rate between the Korean Won and Japanese Yen **in the long run.**